Intent Evals Before and After ArmorIQ

The measured drift baseline, and the projected drift once each task's intent contract is enforced as an Armor policy that prevents an agent from touching anything it was not asked to.

Deliverable 3 of 3.

The before columns are measured live from db/results.db (479 runs). The after columns are PROJECTIONS, derived from the policy coverage in armor-intent/policies, not from Armor runs. The current Cursor quota caps five of seven arms until 30 September 2026, so the enforced re-run that would replace these projections with measurements is deferred, not assumed. No projected number is presented as measured.

1. What Armor enforces, and why the rungs should flatten

The benchmark finding is a prompt-rung effect: with a scoped prompt agents hold, with a loose prompt they drift, with a vague prompt they hold again. That variance lives entirely on the agent side, in how prompt phrasing invites scope expansion. Armor removes the agent's ability to act on that invitation. Its PreToolUse gate refuses any edit to a forbidden file before it lands, regardless of how the prompt was worded, so the drift a loose prompt would have produced never reaches the repository. The scoped and vague rungs are already at zero drift and stay there; the loose rung is where enforcement does its work, and the projection is that it collapses toward the same floor.

2. Loose-rung drift, measured before vs projected after

Each cohort is the loose rung, where all drift lives. Before is the measured drift rate. After (expected) applies the 3-to-1 cross-to-same coverage; after (worst case) is the structural ceiling in which every drifted cell happened to hit the one same-file alias a path policy cannot block. The true measured value will fall between them and is pinned by persisting per-symbol identity.

Cohort (loose)	Before: drift	After: expected	After: worst case
Generation 1, loose	33 / 56 (59%)	~ 8 / 56 (14%)	≤ 33 / 56 (59%)
Generation 2, loose	25 / 49 (51%)	~ 6 / 49 (12%)	≤ 25 / 49 (51%)
New domains, loose (2 arms)	14 / 20 (70%)	~ 4 / 20 (20%)	≤ 14 / 20 (70%)

Scoped and vague rungs, both cohorts: 0 before, 0 after (nothing to prevent). The expected after-rate is a projection, not a measurement.

3. Two enforcement tiers, and where each lands

Path-level policy (shipped Armor conditions) blocks the 51 cross-file rabbit holes of 68 deterministically, 75% of the surface. Its residual is the same-file old_<fn> alias, at most one per cell. Symbol-level enforcement adds Armor's signed intent plan: an edit to the fix file that targets the alias rather than the declared symbol is a tool-not-in-plan denial, which closes the residual and drives covered drift to zero. Armor has both mechanisms; the path tier is a configuration, the symbol tier uses the plan granularity Armor already supports.

Tier	Mechanism	Covered surface	Determinism
Path policy	file.path forbid on rabbit-hole files	51 of 68 holes (75%)	deterministic on covered surface
+ Symbol plan	signed intent plan, tool-not-in-plan	adds the same-file alias residue	deterministic toward 100%

4. Does the fix still get made under enforcement

Preventing drift is only useful if the requested fix still lands. The policy permits edits to the fix file, so the in-scope change is never blocked; enforcement only denies the out-of-scope edits. In enforce mode the projected outcome on a covered cell is the target state the benchmark defines as fidelity: visible and hidden tests pass and unrequested_changes is zero. The honest risk is an agent that, denied a rabbit-hole edit, fails to complete the real fix or loops retrying the blocked call. That is a behavioural question a projection cannot answer; it is the first thing the enforced re-run measures, per arm, as a task-success rate under policy.

5. Data precision achievable now, without live Armor runs

Two things are already exact today. The policy coverage is exact: 51 of 68 holes are cross-file across all 17 domains, a static fact of the fixtures, not a sample. And the structural floor is exact: at least (drift symbols minus drifted cells) edits are provably cross-file and would be denied. What is not yet exact is the split of measured drift between cross-file and same-file, because the scorer stored only the count of changed symbols, not their identity. Recovering that identity is the one missing measurement, and the AST diff already computes it at scoring time, so a single stored column turns every projected band in this document into a measured number, before any Armor run and before the quota resets.

Until then the claims stated as measured are: the before drift rates, the 75% path coverage, and the per-cohort guaranteed floor. Everything about the after state is a projection carrying an explicit band, and the enforced re-run after the 30 September 2026 quota reset is what converts it.